FLEET TEAM DIGEST

737MAX-FTD-34-19005

Issue Title : Honeywell RDR-4000 Weather Radar System Component Reliability

Airplane Model	ATA	Minor Model(s)
737MAX	3441-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
03/05/2019	11/22/2023	04/15/2023

Estimated Completion Date	Next Update Date

Revision Description

Updated Status, Final Action, and Milestone sections and closed FTD with all issue actions complete.

Applicability

737 MAX airplanes with Honeywell RDR-4000 Weather Radar System installed.

Description

737NG/MAX operators have experienced Honeywell RDR-4000 Weather Radar component removals early into service. Honeywell has identified internal issues with the Transmitter-Receiver (TR) P/N 930-2000-010 and Radar Processor (RP) units that have caused the system to fail.

Background

The TR failures have occurred due to the following:

-MOSFET transistor used on the RF Circuit Card Assembly (CCA) overheating and failing to dissipate heat to the solder pad under the part.

-Capacitor cracking has been seen on the RF XMIT CCA and the Controller CCA which use a number of these capacitors

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manufactured by Plexus.

-Application Boot Code software has sometimes been erased from the TR Controller CCA memory devices.

The RP failures have occurred due to the following:

-Solder cracking has been seen on the FL1 48 MHz band pass filter on IF Converter CCA, pn 722-4934-002 which causes the unit to fail.

Status

-**Capacitor cracking**: Findings indicate that the issue was caused by circuit board flexing which caused micro cracks in the capacitors. Corrective action involved modifying CCA tooling and changes to the card test process for both the RF and Controller CCA's. Corrective action was implemented in October 2019.

-**MOSFET failure**: Honeywell reviewed the solder process, and conducted cross sectioning on new boards to determine failure modes. It was determined that MOSFET's (Q101 component) are failing due to the difference in temperature experienced during normal operation.

The following changes will be incorporated into the WxR Transmitter/Receiver and Radar Processor (RP) designs to improve system reliability:

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---A Hardware change is being implemented in the TR design to incorporate a larger MOSFET component with better heat dissipation, increase the size of the CCA thermal drain pad to further dissipate heat, and change the solder material to one with higher temperature resistance.

---A Software change to the RP will also be incorporated to enhance the monitor and control of MOSFET operation, and to keep temperature variations within specification limits.

Document Releases as of 17 August 2021:

---Honeywell Service Information Letter (SIL) D202101001571 detailing the TR-1 unit failures was released 18 January 2021.

---Honeywell Service Bulletin (SB) 930-1000-34-0022 (Publication Number D202012001499) - RP-1 software fix to improve TR performance - dated 15 February 2021.

---Honeywell Service Bulletin (SB) 930-2000-34-0010 (Publication Number D202101001585) - TR-1 hardware fix to increase TR reliability - dated 25 Jun 2021.

-**Boot Code erase:** Identified the source of the problem with 4 separate issues that will need to be corrected with boot code programing changes, and a new image with the sector lock drivers updated to the flash memory standard. An alternate method to deploy the sector locking has been tested in the CCA. The factory can program the flash memory to lock protect two key sectors before shipment: 1) the boot image, and 2) the factory encoded data unique to each LRU, while leaving all of the other sectors unlocked for access by the TRP flight software and boot image. This means the 996-4030-506 Application software does not need to be changed and recertified. A hardware and software fix is being implemented and is expected to be a part of the new transmitter module design.

-**Band Pass Filter cracking:** If the center conductor leads and the grounding tabs of the IF Converter are not perfectly flush with the printed circuit board during the
soldering process. it can lead to poor solderability and eventually cracking due to insufficient solder on the leads and grounding tabs.

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Honeywell was examining and resoldering the center conductor and grounding tab leads of the filter if they found an RP-1 that came to the shop for a weather painting problem or with fault codes 3049 and 3057 stored in the RP-1's fault memory.

Currently, any IF Converter CCAs that are returned for the weather painting problem or with fault codes 3049 and 3057 stored in the RP-1's fault memory will have the IF Convertor CCA repaired or replaced with a new CCA that is considered a drop-in replacement for the faulty cards and does not have the solder issue.

Interim Action

-Honeywell has released a software update to the Radar Processor that enhances fault monitoring and logging capability. This minor software version change does not increment the RP part number and can be incorporated on-wing via Honeywell Service Bulletin 930-1000-34-0020 (Publication Number D201811000016).

Final Action

-Honeywell has released a software update to the Radar Processor that improves TR operation and reliability. This minor software version change does not increment the RP part number and can be incorporated on-wing via Honeywell Service Bulletin 930-1000-34-0022 (Publication Number D202012001499).

-Honeywell has released a hardware update to the Transmitter-Receiver that improves TR operation and reliability. This minor hardware change (mod dot) does not increment the TR part number but requires return of unit to an authorized Honeywell Service Center. Reference Honeywell Service Bulletin 930-2000-34-0010 (Publication Number D202101001585).

-Boot Code erase corrective action - Corrective action will be both hardware and software based to implement the latest memory device with the correct flash memory drivers. To be implemented as part of the Transmitter CCA redesign.

-Band Pass Filter cracking corrective action - The FL1 filter is being changed to a new filter that will ensure good solderability because of the new type of filter center conductor and grounding contacts which are located on the underside of the component. The new CCA is a drop-in replacement for the old and any units sent in for this issue will have the FL1 solder joints inspected for cracks or insufficient solder. If an issue is discovered during inspection that cannot be corrected by resoldering, the RP will get the new CCA as a drop-in replacement part.

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Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Processor Software Service Bulletin D201811000016 available: April 2019

Capacitor root cause/corrective action implemented: October 2019

MOSFET failure root cause established: February 2020

RP Software Service Bulletin available: 15 February 2021

MOSFET CCA lab testing complete: May 2021

MOSFET Production Development, Fabrication, and Implementation changes complete: June 2021

TR Hardware Service Bulletin available: July 2021

Boot Code memory lab testing complete and root cause established: April 2022

Band Pass Filter hardware change identified: July 2022

Band Pass Filter hardware change available for production/retrofit: December 2022

Boot Code memory hardware/software change identified: December 2022

Boot Code memory hardware/software design implementation details: April 2023

Operator Action

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Contact Honeywell regarding incorporation of WxR Processor and Transmitter-Receiver hardware/software updates available via Service Bulletin.

Part Information

WxR Transmitter-Receiver - P/N 930-2000-010

WxR Processor - P/N 930-1000-002 / -003

Supplier Information

Honeywell International Inc.

15001 N.E. 36 Street

Redmond, Washington 98052-5317

Telephone: (800) 601-3099 (Toll Free U.S.A./Canada)

Telephone: (602) 365-3099 (International Direct)

Web site: http://portal.honeywell.com/wps/protal/aero

Honeywell Aerospace Technical Support

Telephone: 855-808-6500 (Toll Free U.S.A./Canada)

Telephone: 602-365-6500 (International Direct)

Option 1 for Avionics or Option 2 for Mechanical

E-mail: aerotechsupport@honeywell.com

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737NG-FTD-34-18002		
Service Bulletin (SB)	930-2000-34-0010 (D202101001585)	0	06/25/2021
Service Information Letter (SIL)	D202101001571	0	01/18/2021
Service Bulletin (SB)	930-1000-34-0022 (D202012001499)	0	02/15/2021
Service Bulletin (SB)	930-1000-34-0020 (D201811000016)	0	05/02/2019

Parts List

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Part Name	Boeing Part Number	Supplier Name	Supplier Number
WxR Transmitter-Receiver (TR)		Honeywell	930-2000-010
WxR Processor (RP)		Honeywell	930-1000-002, -003

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- CRC - Component Reliability Collaboration : Collaborative process to eliminate component removals in production and the during the first year of new airplane in service
- Manufacturing Quality Issues : Issues related to airplane or component manufacturing quality

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